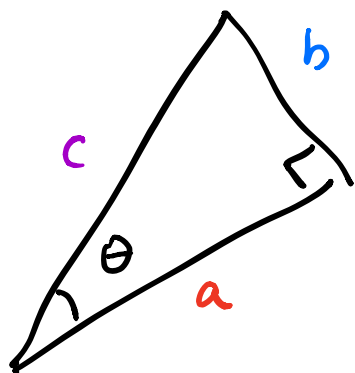


Trigonometry may be the study of triangles, but it relates to so much more, such as the position on a circle, the behavior of a wave, complex numbers, and the relationships between vectors, especially vectors.

There are three trigonometric functions, all defined as ratios of sides of a right triangle.

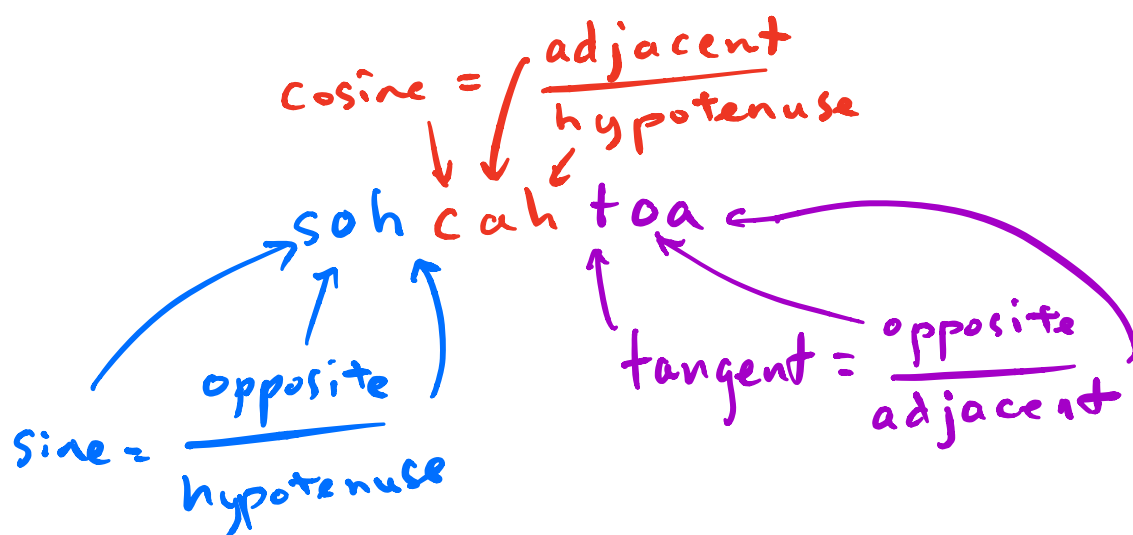


$$\cos \theta = \frac{a}{c}$$

$$\sin \theta = \frac{b}{c}$$

$$\tan \theta = \frac{b}{a} = \frac{\sin \theta}{\cos \theta}$$

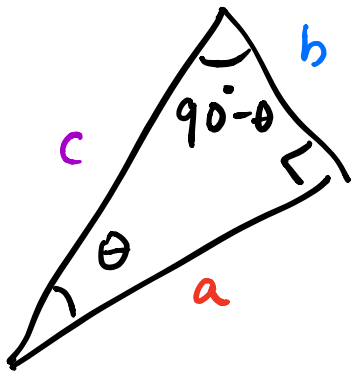
if you need help remembering them,



I always remember them in reference to a circle, but we'll get to that later.

sine and cosine can be related in a couple ways

1)



The sum of angles in a triangle always equals 180° . Hence in a right triangle, the angle opposite θ must be $90^\circ - \theta$.

$$90^\circ + \theta + (90^\circ - \theta) = 180^\circ$$

This allows us to write

$$\cos \theta = \frac{a}{c} = \sin(90^\circ - \theta)$$

and

$$\sin \theta = \frac{b}{c} = \cos(90^\circ - \theta)$$

both very useful relations.

2) Remember the Pythagorean Theorem?

We can relate the length of the sides to the hypotenuse by

$$a^2 + b^2 = c^2.$$

If we divide both sides by c^2 ,

$$\frac{a^2}{c^2} + \frac{b^2}{c^2} = 1.$$

$$\left(\frac{a}{c}\right)^2 + \left(\frac{b}{c}\right)^2 = 1.$$

↑
 $\cos \theta$

↑
 $\sin \theta$

$$\cos^2 \theta + \sin^2 \theta = 1.$$

also

$$\cos^2 = 1 - \sin^2 \theta$$

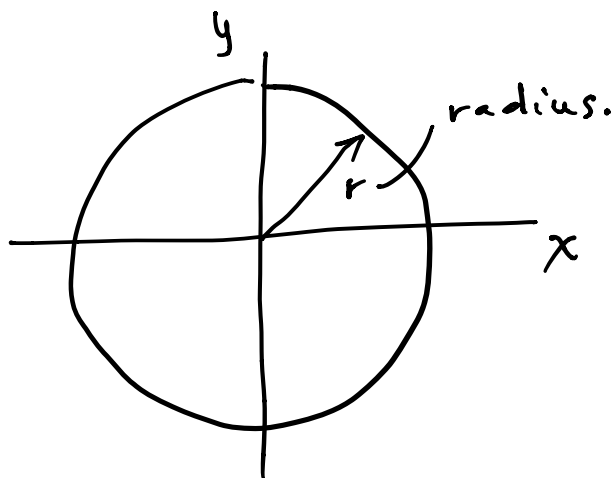
$$\sin^2 \theta = 1 - \cos^2 \theta$$



This identity is amazingly useful and we'll use it several times in the course.

This Pythagorean identity also tells us that we can relate sine and cosine to a circle which has the equation

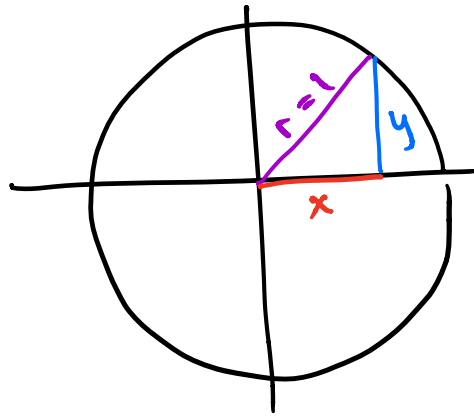
$$\left(\frac{x}{r}\right)^2 + \left(\frac{y}{r}\right)^2 = 1$$



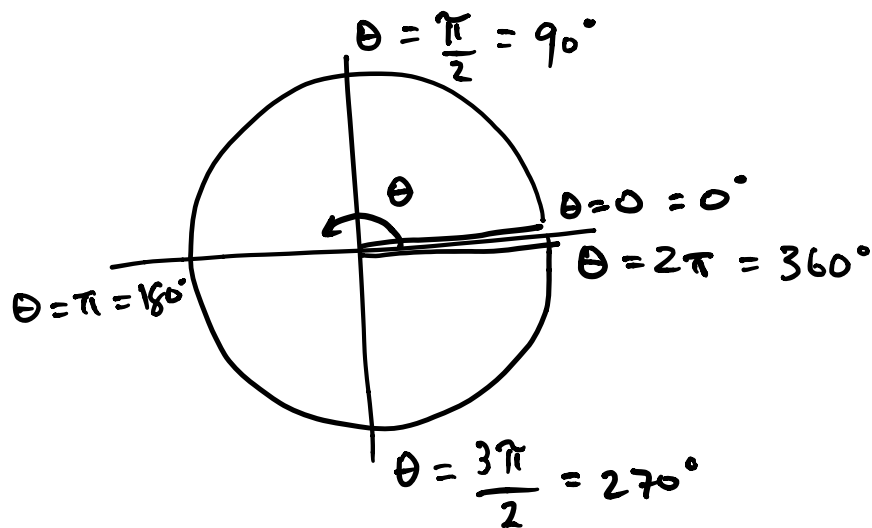
If $r=1$, this is called the unit circle.

$$x = \cos \theta$$

$$y = \sin \theta$$



However, when we use the unit circle, it makes more sense to relate the angles to the circumference. Recall that circumference is $2\pi r$ which for a unit circle is simply 2π .



Other common angles

$$45^\circ = \frac{\pi}{4}$$

$$30^\circ = \frac{\pi}{6}$$

$$60^\circ = \frac{\pi}{3}$$

Since $\cos \theta = x$ and $\sin \theta = y$

$$\cos 0 = 1$$

$$\sin 0 = 0$$

$$\cos \frac{\pi}{2} = 0$$

$$\sin \frac{\pi}{2} = 1$$

$$\cos \pi = -1$$

$$\sin \pi = 0$$

$$\cos \frac{3\pi}{2} = 0$$

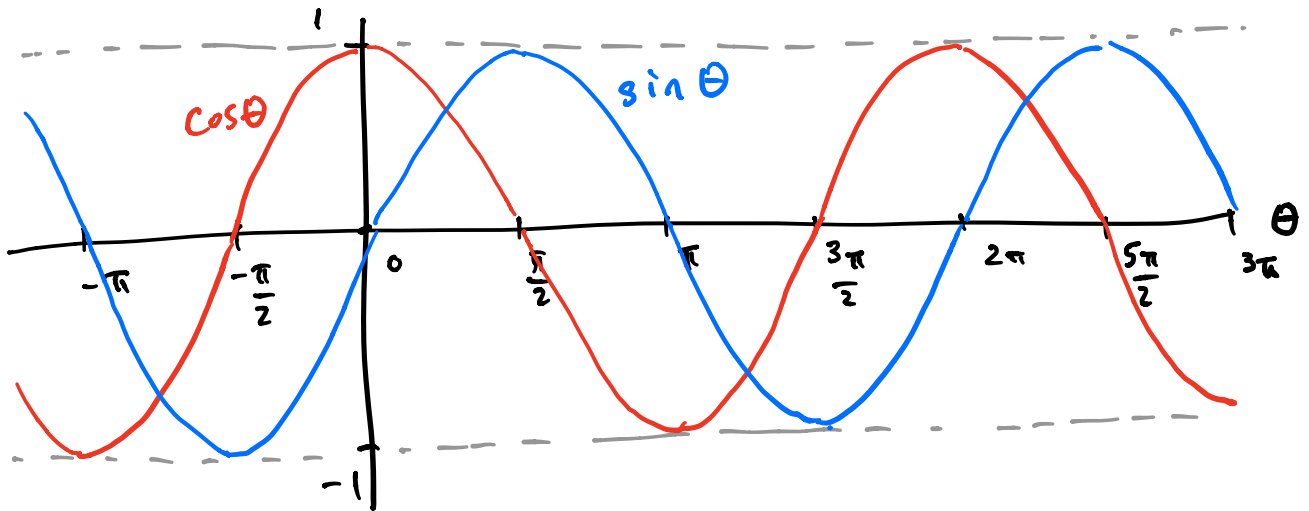
$$\sin \frac{3\pi}{2} = -1$$

$$\cos 2\pi = 1$$

$$\sin 2\pi = 0$$

If we went around the circle again we would see the functions repeat, which describes these trigonometric functions as periodic functions ...

Waves!



Just to wrap-up. Trigonometric functions are more useful than just describing right triangles with some desirable properties that can be exploited finding many applications in geophysics.

They are so useful we will see them again used to build functions with sines and/or cosines. And they will be used to express complex numbers and simplify the math used in electromagnetics.